

Wafer-scale process-decoupled spectral stabilization in SiN CWDM (de)multiplexer

Chih-Hsien Chen⁽¹⁾ and Yung-Jr Hung^(1,*)

⁽¹⁾ Department of Photonics, National Sun Yat-sen University, Taiwan (*yungjr@mail.nsysu.edu.tw)

Abstract We demonstrate wafer-scale process decoupling and spectral stabilization in SiN CWDM photonics using sensitivity-engineered cascaded MZIs. Independent extraction of linewidth and thickness variations enables a $\Delta\lambda/\Delta w$ -compensated design achieving -1.92 ± 0.95 nm wavelength shift with <2 dB insertion loss across a 12-inch wafer.

Introduction

The capacity of optical transceivers is rapidly scaling toward 3.2 Tbps and beyond to meet the explosive bandwidth demand in AI data centers, driving the need for wafer-scale spectrally stable coarse wavelength division (de)multiplexing (CWDM) photonic integration. Silicon nitride (SiN) has emerged as a promising platform due to its low loss and absence of two-photon absorption. However, unlike SOI formed by wafer bonding, chemical vapor deposition (CVD) SiN films exhibit non-negligible thickness non-uniformity across 12-inch wafers. Combined with lithography-induced linewidth variation, these process deviations perturb the effective index and accumulate spectral shifts in cascaded Mach-Zehnder interferometers (MZIs). Although fabrication-tolerant MZI architectures have been reported [1,2] and low-loss SiN CWDM devices with <2 dB insertion loss and <-30 dB crosstalk have been demonstrated [3], wafer-scale experimental decoupling of linewidth (Δw) and thickness (ΔH) variations has not been achieved. Here, we report a sensitivity-engineered SiN CWDM demultiplexer enabling independent extraction of Δw and ΔH across a 12-inch wafer. By implementing a $\Delta\lambda/\Delta w$ -compensated design, we demonstrate a spectral shift of only 1.9 ± 0.9 nm while maintaining high optical performance, establishing a practical design-to-foundry process stabilization methodology for large-scale SiN photonics.

Design and Experiments

The effective index of the SiN waveguide is simulated as a function of waveguide width and thickness, as shown in Fig. 1(a). The results indicate that $\partial n_{\text{eff}}/\partial w$ decreases with increasing waveguide width, whereas $\partial n_{\text{eff}}/\partial H$ increases for wider SiN waveguides. These opposite trends imply that width- and thickness-induced wavelength variations cannot be simultaneously minimized through a single waveguide design

parameter. As shown in Fig. 1(b), a SiN MZI employing a single-mode delayed waveguide ($w = 800$ nm) exhibits wavelength sensitivities of $\Delta\lambda/\Delta w = 0.121$ and $\Delta\lambda/\Delta H = 0.295$. In contrast, a multimode delayed waveguide ($w = 2000$ nm) suppresses $\Delta\lambda/\Delta w$ to 0.015, but at the expense of an increased $\Delta\lambda/\Delta H$ of 0.321, highlighting the intrinsic trade-off between linewidth and thickness sensitivities. These results underscore the necessity of a systematic process evaluation methodology to decouple width and thickness variations and to quantitatively assess their impact on MZI spectral stability across different fabrication platforms. A sensitivity-engineered MZI (SE-MZI) with a central wavelength λ_0 and a defined free spectral range (FSR), as illustrated in Fig. 1(c), can be achieved by simultaneously satisfying Eq. (1) and Eq. (2).

$$\text{FSR} = \lambda_0^2 (n_{g,1} L_1 - n_{g,2} L_2)^{-1} \quad \text{Eq. (1)}$$

$$\frac{\partial n_{\text{eff},1}}{\partial w} L_1 - \frac{\partial n_{\text{eff},2}}{\partial w} L_2 = A \quad \text{Eq. (2)}$$

in which $n_{\text{eff},1}$, $n_{g,1}$, L_1 , $n_{\text{eff},2}$, $n_{g,2}$, L_2 are the effective index, group index and length of two arms in a MZI, respectively, and m is an integer. A determines the linewidth sensitivity. Such a MZI is implemented by tailoring the width-dependent effective index ($\partial n_{\text{eff}}/\partial w$) of the interfering arms through systematic optimization of the waveguide widths and lengths. The absolute wavelength shift of a MZI with respect to width and thickness variation are given by Eq. (3).

Fig. 2(a) presents the wavelength sensitivities ($\Delta\lambda/\Delta w$ and $\Delta\lambda/\Delta H$) of a SiN MZI as a function of the design parameter A . It is observed that $\Delta\lambda/\Delta w = 0$ at $A = 0$, whereas $\Delta\lambda/\Delta H$ vanishes at $A \cong 30$, highlighting the opposite dependence of linewidth and thickness sensitivities on A . The $\Delta\lambda/\Delta w$ -compensated SiN MZI exhibits wavelength sensitivities of $\Delta\lambda/\Delta w = 0.001$ and $\Delta\lambda/\Delta H$

$$\Delta\lambda = \lambda_0 \left[\Delta w \left(\frac{\partial n_{\text{eff},1}}{\partial w} L_1 - \frac{\partial n_{\text{eff},2}}{\partial w} L_2 \right) + \Delta H \left(\frac{\partial n_{\text{eff},1}}{\partial H} L_1 - \frac{\partial n_{\text{eff},2}}{\partial H} L_2 \right) \right] / (n_{g,1} L_1 - n_{g,2} L_2) \quad \text{Eq. (3)}$$

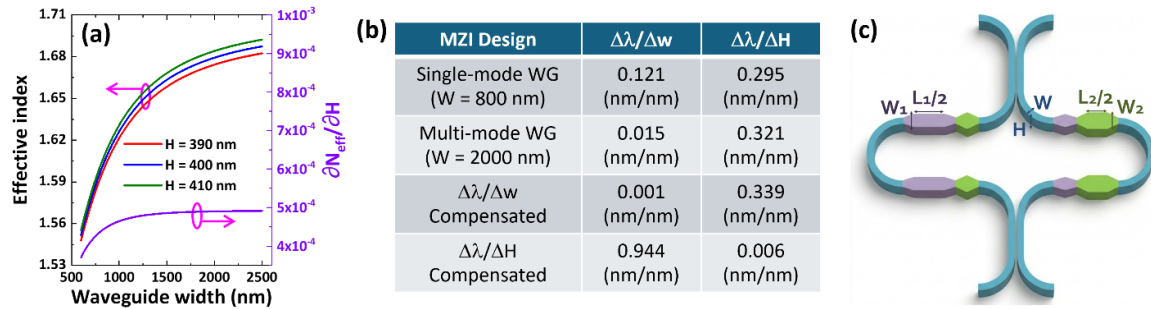


Fig. 1: (a) Simulated effective index of a SiN waveguide as a function of waveguide width and thickness. The sensitivity $\partial n_{\text{eff}}/\partial H$ across different waveguide widths is also analyzed. (b) Extracted wavelength sensitivities ($\Delta\lambda/\Delta w$ and $\Delta\lambda/\Delta H$) of a SiN MZI under different delayed-waveguide designs: single-mode ($w = 800$ nm), multimode ($w = 2000$ nm), $\Delta\lambda/\Delta w$ -compensated, and $\Delta\lambda/\Delta H$ -compensated configurations. (c) Schematic illustration of a fabrication-tolerant MZI.

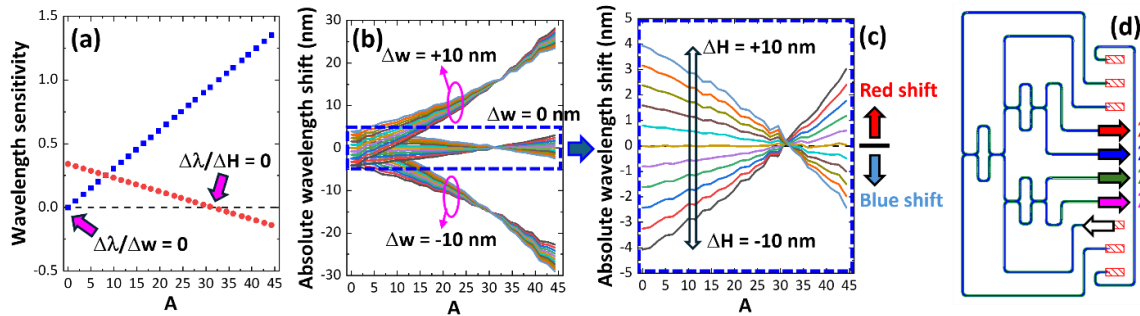


Fig. 2: (a) Wavelength sensitivities ($\Delta\lambda/\Delta w$ and $\Delta\lambda/\Delta H$) of a SiN MZI under different A values. (b,c) Absolute wavelength shifts of the SiN MZI under different A values within ± 10 nm variations in waveguide width and/or thickness. (d) The schematic illustration of a four-channel cascaded SiN MZI.

= 0.339, while the $\Delta\lambda/\Delta H$ -compensated counterpart shows $\Delta\lambda/\Delta w = 0.944$ and $\Delta\lambda/\Delta H = 0.006$, consistent with the data shown in Fig 1(b). The absolute wavelength shifts of the SiN SE-MZIs under different A values within ± 10 nm variations in waveguide width and/or thickness are presented in Fig. 2(b) and 2(c). The results clearly indicate that width variations induce a larger absolute wavelength shift than thickness variations in the SiN MZI. To experimentally validate the sensitivity-engineering concept, four-channel cascaded SiN SE-MZI circuits spanning 28 distinct A values ranging from 0 to 45 were designed and fabricated, as illustrated in Fig. 2(d). The target channel wavelengths were set at 1270, 1290, 1310, and 1330 nm. All directional couplers are implemented as 3-dB power splitters, incorporating low-loss waveguide bend designs [3].

Fig. 3(a) presents the measured spectral responses of the SiN cascaded SE-MZIs with $A = 0, 8.23, 32.82$, and 44.27 at the 1310 nm optical channel. All MZI circuits were designed to exhibit their spectral peaks at 1310 nm; however, deviations are observed due to variations in the fabricated waveguide width and thickness. For the case of $A = 44.27$ a pronounced blue shift of approximately 20 nm is observed. The spectra shown in Fig. 3(a) were measured from devices on a single die, while the identical mask layout was replicated across a 12-inch wafer, resulting in multiple dies. Fig. 3(b) summarizes the

measured spectral responses for all optical channels across 28 SiN cascaded SE-MZIs obtained from five dies located at both the center and edge of the wafer. All channels exhibit a consistent blue-shift trend as a function of A. The absolute wavelength shifts at the 1310 nm channel were extracted for all devices and fitted to Eq. (3) to determine the mean values and standard deviations of the as-fabricated SiN waveguide width and thickness. Emphasis was placed on $A = 0$ ($\Delta\lambda/\Delta w = 0$) and $A \approx 30$ ($\Delta\lambda/\Delta H = 0$), corresponding to width- and thickness-insensitive designs, respectively. The fitting results indicate a linewidth deviation of -8.5 ± 1.5 nm and a thickness deviation of -3 ± 3 nm across the five measured dies.

Finally, SiN CWDM wavelength (de)multiplexers based on cascaded SE-MZIs was designed with a footprint of $\sim 1.8 \times 2.1$ mm. We applied cross-port connection scheme to achieve broadband operation without requiring broadband directional couplers [4]. The proposed (de)multiplexer circuit was fabricated based on a CMOS-compatible process with 193-nm deep ultraviolet lithography provided by a foundry. The photograph of as-fabricated device is shown in Fig. 4(a). The same (de)multiplexer circuit with single mode ($w = 800$ nm) SiN waveguides was also fabricated for comparison. All fabricated devices exhibit an insertion loss of < 2 dB and an optical

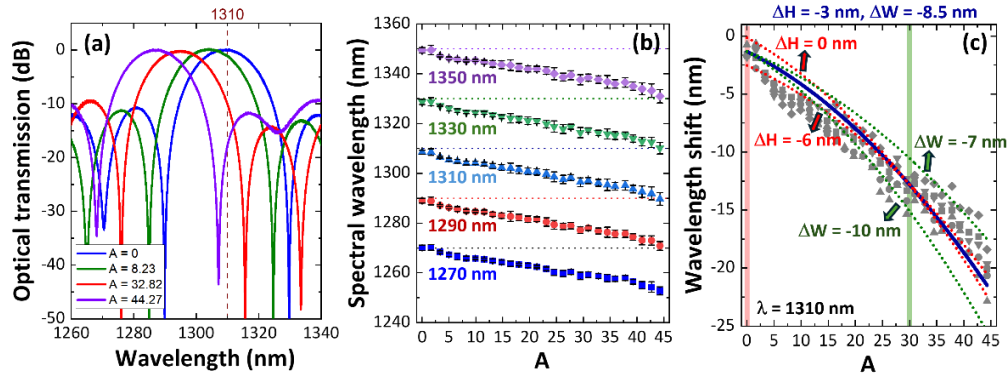


Fig. 3: (a) Measured spectral responses of the SiN SE-MZIs with $A = 0, 8.23, 32.82,$ and 44.27 at the 1310 nm optical channel. (b) Measured channel wavelengths for all four channels across 28 SiN SE-MZIs. (c) Extracted wavelength shifts at the 1310 nm channel for the 28 SiN SE-MZIs, together with fitting to Eq. (3) to determine the mean value and standard deviations of as-fabricated SiN waveguide width and thickness.

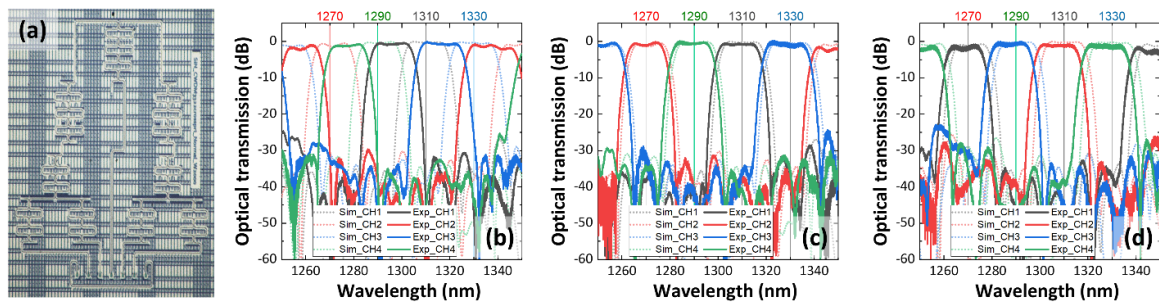


Fig. 4: (a) Photograph of a SiN CWDM wavelength (de)multiplexer based on cascaded SE-MZIs, and their (b-d) Simulated and measured spectral responses: (b) $\Delta\lambda/\Delta H$ -compensated, (c) $\Delta\lambda/\Delta w$ -compensated, (d) single-mode ($w = 800$ nm) configurations.

Tab. 1: State-of-the-art comparison of on-chip CWDM (de)multiplexer

Demultiplexer design	Channel spacing (nm)	Inter-channel crosstalk (dB)	Excess loss (dB)	Wavelength shift (nm)	Reference
Si MZIs (DC)	20	< -33	-5.4	0.91 ± 0.47	[4]
Si MZI (DC)	20	< -19	-6	2	[5]
Si MZI (MMI)	20	< -21	-2.34	3.76	[6]
SiN MZI (DC)	20	< -13	-0.72	0.83 ± 1.7	[7]
SiN MZIs (DC)	20	< -29	-2	1.92 ± 0.95	This work

crosstalk of < -30 dB, as shown in Fig. 4(b-d). The measured wavelength shifts in the $\Delta\lambda/\Delta w$ -compensated and $\Delta\lambda/\Delta H$ -compensated SiN cascaded SE-MZIs were -1.92 ± 0.95 and -14.13 ± 2.72 nm, respectively. For comparison, SiN cascaded MZIs with single mode waveguide exhibits a wavelength shift of -3.06 ± 0.75 nm. Tab. 1 summaries state-of-the-art CWDM (de)multiplexers based on Si and SiN platforms. While SiN devices generally offer low loss, they are more susceptible to spectral variations due to process non-uniformity. In contrast, this work demonstrates improved spectral stability while maintaining competitive optical performance, highlighting the effectiveness of the proposed sensitivity engineered design.

Conclusions

We experimentally demonstrated wafer-scale

process-decoupled spectral stabilization in SiN CWDM (de)multiplexer using cascaded SE-MZIs. The proposed approach enables independent extraction of linewidth and thickness variations, confirming linewidth variation as the dominant contributor to spectral drift across the wafer. The $\Delta\lambda/\Delta w$ -compensated CWDM (de)multiplexer design achieves improved spectral stability while maintaining low loss and low crosstalk. Compared with prior Si and SiN CWDM (de)multiplexers summarized in Tab. 1, this work demonstrates a favorable balance between spectral stability and optical performance, highlighting its potential for wafer-scale integration. This methodology is scalable to higher channel counts and more complex (de)multiplexing architectures, establishing a practical bridge between device design and foundry-level process control for large-scale SiN photonic integration.

Acknowledgments

The authors would like to acknowledge photonic layout support by Synopsys Inc. and chip fabrication support by TSMC. This work was supported in part by the Joint Development Project between TSMC and NSYSU.

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